LOGARITHMS

Common and natural logarithm graphs

Vertical stretches of each other.

Common logarithm

$y = \log(x)$

Through $(1, 0)$ and $(10, 1)$

USE FOR decimal scales and measurements

CONSTANT RATIO, ABOUT 2.303

$$\ln(x) = \ln(10) \cdot \log(x)$$

Natural logarithm

$y = \ln(x)$

Through $(1, 0)$ and $(e, 1) \approx (2.718, 1)$; steeper for small x

USE FOR continuous growth, calculus

SHARED SHAPE

Both have a vertical asymptote at $x = 0$ and extend to $+\infty$; for $x > 1$, $\ln(x) > \log(x)$ because the natural logarithm uses the smaller base. Converting $\log(100) = 2$ multiplies by 2.303: $\ln(100) \approx 4.605$.